Student Registration System

Lab Project Specification & Checklist for Web Development

PROJECT OBJECTIVE

Develop a client-side web application using HTML, CSS, and JavaScript. The system must collect student registration data, perform real-time field validation using Regular Expressions (Regex), display clear visual feedback, and dynamically update a record table upon valid form submission without page reloads.

SYSTEM REQUIREMENTS

1. HTML5 Structure

- **Registration Form:** Form fields for Student ID, Full Name, Email, Phone Number, and Password.
- **Error Messages:** Inline feedback containers (e.g., `<small>`) below each input field.
- **Controls:** Primary *Submit* button and secondary *Reset* button.
- **Display Table:** Semantic `<table>` with columns for ID, Name, Email, and Phone.

2. CSS3 Layout & Visuals

- **Container Layout:** Centered responsive card design with clean padding and distinct sections.
- **Valid State:** Highlight inputs green with light tint when Regex validation passes.
- **Invalid State:** Highlight inputs red with error text when validation fails or field is missing.
- **Button Styling:** Interactive hover states for submission and reset actions.

3. Regex Validation Rules

- **Student ID:** Specific institutional code format (e.g., `CSE-2026-001`).
- **Full Name:** Alphabetic characters and spaces only (2–40 chars).
- **Email:** Standard pattern (`user@domain.ext`).
- **Phone:** Regional 11-digit numeric format starting with 01.
- **Password:** Minimum 8 chars with combined letters and digits.

4. JavaScript & DOM Logic

- **Real-Time Listener:** Run validation on input event as user types.
- **Prevent Default:** Intercept submission to keep application on single page.
- **DOM Appending:** Construct a new row (`<tr>`) and insert valid data into table body.
- **State Reset:** Clear inputs and remove validation classes upon submit or reset.

VERIFICATION & TESTING CHECKLIST

#	Task Requirement	Expected Outcome
1	Empty Form Submission	Prevents submission, marks required fields red, displays warnings.
2	Real-time Validation	Toggles input highlight from red to green instantly upon typing valid data.
3	Dynamic Data Insertion	Appends new student row to the table without refreshing the web page.
4	Form Reset Action	Clears all form controls, removes error messages, and resets field states.